

Danilov Sergey Dmitrievich

Saint Petersburg, Russia · +7 (986) 767-86-96 @geniusserg
danilovsergeyd@vk.com linkedin.com/in/geniusserg

> ML/LLMOps Engineer who owns the full lifecycle of production ML and LLM serving — model deployment, distributed inference, GPU orchestration, vLLM and VeRL, batching and scheduling, MoE routing, and predictive reliability of cluster hardware (memory, network) — turning models into reliable, high-availability services with measurable SLO.

CAREER

Huawei — LLM Inference R&D Engineer · *September 2024 – present*

- Disaggregated Prefill/Decode simulator — C++ discrete-event simulation of a prefill/decode serving system with KV-cache-aware load balancing and Markov request traffic; selects TP/DP/SP/EP topology and scheduling policy to hold 99.99% SLO, and steers vLLM recovery for RL rollouts.
- Time-series clustering of optical-power telemetry flags degrading links early; a minimal-parameter model that raised cluster availability by +10%.
- NUMA-aware stress testing reached 95% of theoretical bandwidth; Weibull MTTR modeling + XGBoost failure forecasting cut repair costs by 2×. doc
- Combinatorial EPLB MoE expert-placement algorithms that co-optimize expert-to-GPU mapping with all-to-all communication to minimize network congestion and tail latency; the same platform was used for multi-node deployment of the optimized EPLB algorithms in vLLM, running benchmarks, and building detailed dashboards.
- Deep knowledge across the serving stack: continuous batching and preemptive scheduling, fused communication operators, FP8/INT8 quantization, speculative decoding, and optimization trade-offs.
- LLMOps for an LLM quantization project: owned the full lifecycle from research notebook → quantized model → deployed service and benchmarks, speeding up the research cycle by 3×.

ML Engineer — ITMO University · *November 2023 – September 2024*

- Sber & ITMO: CTGAN user emulator; improved BiVAE recommendation MAP@5 by 3%.
- Almazov & ITMO: Shapley-valued weighted XGBoost for obesity treatment; 71% accuracy for 3-month sibutramine therapy response; 2 Q2 papers. J. Pers. Med. J. Clin. Med.
- Industrial AI assistant: improved PDF NER with prompt engineering/self-consistency; filtering removed 80% repetition.

Testing Engineer — Harman · *September 2021 – July 2022*

- Sped up testing for complex GPS automotive systems with Robot Framework

Release Engineer — Intel · *July 2019 – August 2021*

- Supported Intel OneAPI HPC release infrastructure: Python/Bash automation and C++ installer.

TECH STACK

- **AI infrastructure:** Harness, Agents workflow, RAG, CAG, vLLM, VeRL, PyTorch, Kubernetes
- **Data Analysis & Research:** Python, Pandas, Scikit-learn, AutoML, PatentScope, Elsevier Scopus

EDUCATION

ITMO University, 2024 — Master's degree in Data Analysis

– Thesis: Shapley-value weighted filtering for small-sample ML; +7% accuracy on medical data.

Higher School of Economics, 2022 — Bachelor's degree in Software Engineering

– Thesis: accessible assistant for visually impaired users using Yandex Cloud OCR.

PUBLICATIONS

- Google Scholar · h-index 2
- Danilov et al., J. Pers. Med. 2024, 14, 811 — sibutramine therapy prediction.
- Markina et al., J. Clin. Med. 2024, 13, 4151 — brown adipose tissue and obesity treatment.

COMPETITIONS

ScienceRAG — Research assistant agent based on GigaChat

– ScienceRAG: improved ArXiv RAG retrieval on GigaChat YouTube

X5 AI Tech Hackathon — Hallucination detection algorithm

– X5 AI Tech: ruBERT hallucination detector — 88% accuracy, <0.45 s inference; top-10.

LANGUAGES

- Russian, English (B1)